MILITARY SPECIFICATION

GYRO, VERTICAL AND MOUNT (TYPE MF-1 COMPASS)

This specification has been approved by the
Bureau of Aeronautics, Department of the Navy.

1. SCOPE

1.1 Scope.- This specification covers design requirements and all performance requirements for procurement of a Miniature Vertical Gyro and Mount, a component of the MF-1 type compass directional reference system for military aircraft.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and publications of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-N-290	Nickel Plating (Electrodeposited)
QQ-C-320	Chromium Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-P-116	Preservation, Method of
MIL-S-4456	Shock, Variable Duration, Method and Apparatus for
MIL-D-70327	Drawings, Engineering and Associated Lists
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-I-6181	Interference Limits, Tests and Design Requirements, Aircraft Electrical and Electronic Equipment
MIL-S-6872	Soldering Process, General Specification for
MIL-L-6880	Lubrication of Aircraft, General Specification for
MIL-S-7742	Screw Threads, Standard Aeronautical
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-A-8625	Anodic Coating for Aluminum and Aluminum Alloys
MIL-C-21013	Compass Directional Reference System Magnetic Slave, Directional Gyro Roll Stabilized, 3°/HR Type MF-1

STANDARDS

Federal

FED. STD. No. 1 Standard for Laboratory Atmospheric Conditions

PSC 6605

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Military

MIL-STD-130	Identification Marking for U.S. Military Property
MIL-STD-200	Electron Tubes and Semiconductor Devices, Diode
MIL-STD-701	Transistors
MS28101	Gyro Vertical, and Mount (Type MF-1 Compass)
MS33586	Metals - Definitions of Dissimilar

PUBLICATIONS

Air Force-Navy Aeronautical Specification Bulletin

No. 143 Specifications and Standards, Use of

(When requesting any of the applicable documents, refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania, Attention: Code CDS.)

3. REQUIREMENTS

3.1 Qualification.- The Vertical Gyro furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein.

3.2 Selection of Government Documents.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with ANA Bulletin No. 143, except as provided in Sections 3.2.1 Commercial Parts and 3.2.2 Standard Parts.

3.2.1 Commercial Parts.- Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, cotter pins, having suitable properties may be used provided:

(a) They can be replaced by the standard parts (MS or AN) without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.2.2 Standard Parts.- With the exception of Section 3.2.1 Commercial Parts, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials.- Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.3.1 Critical Materials.- Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 Metals.- Metals shall be of the corrosion-resistant type, or shall be suitably protected as specified herein to resist corrosion due to salt spray or atmospheric conditions to which the Vertical Gyro may be subjected when in storage or during normal service life.

3.3.2.1 Dissimilar Metals.- Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in Standard MS33586 shall not be used in intimate contact with each other.

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3.3.3 Nonmagnetic Materials.- Nonmagnetic materials shall be used for all parts of the Vertical Gyro, except where magnetic materials are essential.

3.3.4 Fungus-proof Materials.- Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.5 Corrosive Fumes.- The materials used in the Vertical Gyro shall not liberate deleterious fumes.

3.4 Component Parts.- The Vertical Gyro shall consist of the following component parts:

(a) Hysteresis Gyro Motor
(b) Erection Circuit
(c) Two Synchro Transmitters
(d) Hermetically Sealed Case
(e) Shock Mount

3.5 Design and Construction.- The Vertical Gyro, covered by this specification, shall provide an electrical representation of displacement angles about the pitch and roll axes. It shall have unlimited mechanical freedom about the roll axis and approximately ±85 degrees of freedom of the pitch element about its axis. The Vertical Gyro shall maintain interchangeability of the Type MF-1 compass system into various aircraft configurations. Means shall be provided for the following:

(a) Electrical representation of displacement angles about the pitch and roll axes.
(b) Unlimited freedom about the roll axis and approximately ±85° of freedom about the pitch axis.
(c) Electrolytic vertical reference and associated erection circuit.
(d) Circuit to stop erection in both pitch and roll axes.
(e) Test circuit to allow check of erection rates.

The erection circuit shall consist of two electrolytic vertical sensors and torque motors. These separate sensors, with their associated relays and torquers, impart individual control to the erection of both pitch and roll gimbals. A separate circuit shall allow the gimbals to be torqued away from the vertical position, either together or separately, so that the erection rates may be checked. The design of the vertical sensors shall be such that the torquing force applied to the gimbals shall be proportional to gimbal displacement from one degree of each side of vertical. Switch design shall be such that internal pressures built up during normal operation will not adversely affect the life expectancy of the equipment.

The Vertical Gyro shall be constructed to withstand the normal strains, jars, vibrations, and such other conditions as are incident to shipping, storage, installation, and service.

3.5.1 Maintenance.- The design of the Vertical Gyro shall be such as to accommodate to the greatest possible extent, disassembly, reassembly, and service maintenance by those tools and items of maintenance equipment which are normally available as commercial standards.

3.6 Performance.- The Vertical Gyro shall perform satisfactorily when subjected to the tests specified in Section 4 Quality Assurance Provisions.

3.7 Case.- The outline dimensions of the Vertical Gyro case shall conform to Standard MS28101.

3.8 Screw Threads.- Screw threads 0.060 inch or larger in diameter shall be in accordance with Specification MIL-S-7742.

3.9 Electronic Parts.- Electronic parts and the application thereof shall be in accordance with Specification MIL-E-5400. Parts that do not appear on approved lists in accordance with Specification MIL-E-5400 shall not be used unless approved by the procuring activity.

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3.9.1 Tube Types.- The choice and application of electron tubes and transistors shall be in accordance with Standard MIL-STD-200 and MIL-STD-701.

3.9.2 Wiring.- The electrical wiring system shall be insulated from the Vertical Gyro case. The external wiring shall be in accordance with the wiring diagram in Specification MIL-C-21013.

3.9.3 External Test Points.- The Vertical Gyro unit shall be provided with the following minimum external test points:

J1-H (Manual Fast Erection)
J1-K (Reverse Erection)
J1-V (Pitch Erection Cutout)
J1-M (Roll Erection Cutout)

3.9.4 Electrical Receptacle.- The electrical receptacle used shall be in accordance with Standard MS28101.

3.9.5 Electrical Power Requirements.- The Vertical Gyro shall operate from a 115 volt, 3 phase, 400 cycle per second power supply.

3.10 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part number shall be governed by the drawing number requirements of Specification MIL-D-70327.

3.11 Physical Requirements.-

3.11.1 Weight.- The weight of the Vertical Gyro and integral shock mount shall not exceed 5.75 pounds.

3.11.2 Finish.- Protective coatings and finishes which will crack, chip, or scale during normal service life or due to extreme of atmospheric conditions shall not be used.

3.11.2.1 Aluminum Alloy Parts.- Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to Specification MIL-A-8625. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with Specification MIL-C-5541.

3.11.2.2 Iron and Steel Parts.- Where practicable, iron and steel parts shall be cadmium, chromium,or nickel plated in accordance with Specifications QQ-P-416, QQ-C-320, and QQ-N-290, respectively.

3.11.3 Soldering.- Soldering shall be performed in accordance with Specification MIL-S-6872.

3.11.4 Lubrication.- Lubrication shall be accomplished in accordance with Specification MIL-L-6880, except that the selection of the lubricant used shall be approved by the procuring activity.

3.12 Marking.- All markings shall be durable to withstand usage encountered in service.

3.12.1 Identification of Product.- A nameplate shall be securely attached to the exterior of Vertical Gyro case, and shall be legibly and durably marked in accordance with Standard MIL-STD-130.

3.12.2 Receptacles.- The electrical receptacle code markings as specified on Standard MS28101 shall be legibly and permanently applied to the Vertical Gyro case.

3.13 Installation.-

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3.13.1 Installation Instructions.- The contractor shall furnish with each Vertical Gyro one printed copy of instructions with illustrations and diagrams, if necessary, covering the installation of the Vertical Gyro. Prior to printing, two copies shall be furnished to the Bureau of Aeronautics for approval. The instructions shall be printed on 8-1/2 by 11 or 11 by 17 inch paper.

3.13.2 Envelope.- An envelope furnished by the contractor containing installation instructions (and mounting screws if required) shall be packaged with the following information:

"IMPORTANT
THIS ENVELOPE CONTAINS INSTRUCTIONS AND MOUNTING SCREWS"

3.14 Workmanship.- The Vertical Gyro including all parts and accessories, shall be constructed and finished to produce an instrument free from all defects which would affect proper functioning in service. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Classification of Tests.- The inspection and testing of the Vertical Gyro shall be classified as follows:

(a) Qualification tests.- Qualification tests are those tests performed on Vertical Gyro submitted for qualification as a satisfactory product.

(b) Acceptance tests.- Acceptance tests are those tests performed on Vertical Gyro manufactured and submitted for acceptance under contract.

4.3 Qualification Tests.- The qualification tests of the Vertical Gyro shall consist of all tests of this specification. The tests shall be conducted in the following sequence:

Individual Tests
Sampling Test
Voltage and Frequency Variation
Radio Noise Interference
High Temperature Exposure
Low Temperature Exposure
Vibration
Impact Shock
Life
Humidity
Salt Spray

4.3.1 Qualification Test Samples.- The qualification test samples shall consist of two Vertical Gyros of each manufacturer's part number submitted as sample for qualification testing. After obtaining authorization for submittal from the Bureau of Aeronautics, the samples shall be forwarded to the testing laboratory designated in letter of authorization. Each sample shall be plainly identified by a securely attached durable tag marked with the following information (see Section 6.3 Qualification):

Sample for qualification test
Submitted by (name) (date) for qualification tests in
accordance with the requirements of Specification MIL-
under authorization (reference letter authorizing test)

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Gyro-Vertical and Mount (Type MF-1 Compass)
MS28101
Manufacturer's Part No.
Name of Manufacturer

4.4 Acceptance Tests.- The acceptance tests of the Vertical Gyro shall consist of the Individual tests and the sampling tests of this specification.

4.4.1 Individual Tests.- The Individual tests of the Vertical Gyro shall consist of the following tests conducted on each Vertical Gyro:

Inspection
Dielectric Strength
Insulation Resistance
Initial Erection
Final Erection
Current Consumption
Synchro Requirements
Erection Rate
Manual Fast Erection
Drift
Vertical Erection Error (Pitch and Roll)
Loop Test
Power Interruption

In addition, each unit may be subjected to any other test specified herein which the inspector considers necessary to determine compliance with the requirements of this specification.

4.4.2 Sampling Tests.- The sampling tests of the Vertical Gyro shall consist of the following tests conducted on each sampling test sample:

System Vertical Gyro Tests
High Temperature Operation
Low Temperature Operation
Immersion (Leakage) Test

4.4.2.1 Sampling Test Instructions.- Samples shall consist of two Vertical Gyros selected at random by the inspector from each lot of 100 or less Vertical Gyros which have passed the individual tests. A lot shall consist of identical Vertical Gyros with the same manufacturer's part number, manufactured under substantially the same conditions, and submitted at substantially the same time. Vertical Gyros which have been subjected to the sampling tests shall not be delivered on contract until they have been rebuilt and resubmitted to all individual tests.

4.5 Rejection and Retest.- Vertical Gyros which have been rejected as a result of the acceptance tests may be reworked to correct the defects, or replaced, and resubmitted for acceptance. Failure of samples representing lots shall be cause for rejection of the lot represented. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Inspector. Units or lots rejected after retest shall not be resubmitted without specific approval of the procuring activity.

4.6 Test Conditions.-

4.6.1 Atmospheric Conditions.- Unless otherwise specified, all tests required by this Specification shall be made in accordance with the requirements of Federal Standard No. 1.

4.6.2 Attitude.- Unless otherwise specified, the units under investigation shall be tested in their normal operating positions with the mounting surface level within six minutes of arc.

4.6.3 Test Equipment.- Any test or metering equipment shall not load or affect the unit under test. The tolerances indicated are acceptable limits of performance and are not to be confused with meter accuracies.

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4.6.4 Altitude.- Whenever the requirements of this specification prescribe a given altitude, a pressure equivalent to the altitude specified, based on the U. S. Standard Atmosphere, shall be employed.

4.6.5 Standard Power Source.- Whenever a standard power source is specified, it shall be 115 ±3 volts, 400 ±10 cycles per second (cps), 3-phase power supply, ABC rotation.

4.6.6 Circuit Connections.-

Pin	Circuit
P	115 volts, 400 cps A phase (Ground)
R	115 volts, 400 cps B phase
S	115 volts, 400 cps C phase
D	Roll Axis Synchro Stator (Blue)
E	Roll Axis Synchro Stator (Yellow)
F	Roll Axis Synchro Stator (Black)
A	Pitch Axis Synchro Stator (Blue)
B	Pitch Axis Synchro Stator (Yellow)
C	Pitch Axis Synchro Stator (Black)
V	Pitch Erection Cutout (Connect to "S")
M	Roll Erection Cutout (Connect to "S")
T, U	Power failure indicating circuit (closed during normal erection and free drift. Open when no power applied, when any of three phases are open, incorrect phase rotation, and when fast erection circuit is closed.)
H	Fast Erection, Both Axes (Connect to "S")
K	Test (Connect to "S" to drive spin axis away from vertical, both axes at normal erection rate.) Connect "H" and "K" to "S" to drive spin axis away from vertical, both axes, at fast erection rate.

4.6.7 Pitch, Roll, and Yaw Motion.- When a pitch, roll, and yaw testing apparatus is specified, it shall provide 1-1/2° through 7-1/2° of angular motion each side of the vertical reference axis and operate at a speed of 5 to 7 cycles per minute and reverse uniformly at least once every minute.

4.6.8 Warm-Up Time.- A ten minute warm-up will be allowed before the conduction of any test unless otherwise stated.

4.7 Test Methods.-

4.7.1 Inspection.- Each Vertical Gyro shall be examined to determine conformance with the applicable drawings and with all the requirements of this specification not covered by tests.

4.7.2 Dielectric Strength.- A potential of 250 volts (root mean square) alternating current at commercial frequency shall be applied between each pin, except V and M, in the electrical connector and the case ground for a period of five seconds. There shall be no breakdown of insulation or any other permanent damage to the instrument as a result of this test.

4.7.3 Insulation Resistance.- A potential of 500 volts direct current shall be applied prior to helium filling between each pin, except V and M, in the electrical connector and the case ground for a period of five seconds. The insulation resistance shall be 50 megohms minimum.

4.7.4 Initial Erection.- With the gyro rotor at rest and the spin axis at any random position within ±80° in roll and pitch, power shall be applied to the unit. Within 60 seconds, the gyro shall indicate vertical within ±1°.

4.7.5 Final Erection.- After the 30 second initial erection period, the Vertical Gyro shall indicate vertical within ±1/2° in a period not to exceed two minutes. The unit shall then be rapidly rotated 180° in azimuth. The pitch and roll outputs shall not indicate a change in verticality of the spin axis by more than 1°.

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4.7.6 Current Consumption.- After ten minutes of operation, the current in each of the three power leads (P, R, S) shall not exceed 0.125 ampere during normal erection. The current shall not exceed 0.250 ampere when fast erection is applied.

4.7.7 Synchro Requirements.-

4.7.7.1 Synchro Null.- With the gyro in operation, and the normal erection voltage applied, the open circuit synchro null voltages of both the pitch and roll axes shall not exceed 0.050 volt rms measured across pins A and B on the pitch axis and D and E on roll axis.

4.7.7.2 Synchro Maximum Output Voltage.- With the gyro operating at rated power and with normal erection voltages applied, measure the maximum take-off output voltage as the Vertical Gyro case is tilted clockwise and counter-clockwise about each axis to achieve maximum outputs for each of the four displacements. The open-circuit take-off output voltage for all four displacements shall be 11.8 volts ±0.6 volt rms across pins A and B for pitch and D and E for roll output.

4.7.7.3 Synchro Zeroing and Sensing.-

(a) Connect circuits C and F to P.
(b) Connect A to B, voltage between this point and R shall be less than 115 volts.
(c) Connect D to E, voltage between this point and R shall be less than 115 volts.
(d) Disconnect jumpers between A and B and between D and E. Voltage between "D" and "F-P" shall increase in right bank before decreasing. Voltage between "B" and "C-P" shall increase in climb before decreasing.
(e) Remove jumper between C and F to P.

4.7.8 Erection Rate.- With the Vertical Gyro case level and with standard power applied, perform the following tests:

4.7.8.1 Pitch Axis.- Connect pins H and K to S and drive the spin axis to approximately 15° climb. (Connect pin M to S to stop the roll axis from deflecting.) Remove pins K, H, and M from S and allow the gyro to erect. The time to erect from 10° to 5° shall be between 1 minute 15 seconds and 2 minutes 30 seconds. Repeat this test for dive. At no time during this test shall the roll axis deviate from null by more than 1°.

4.7.8.2 Roll Axis.- Connect pins H and K to S and drive the spin axis to approximately 15° right bank. (Connect pin V to S to stop the pitch axis from deflecting.) Remove pins K, H, and V from S and allow the gyro to erect. The time to erect from 10° to 5° shall be between 1 minute 15 seconds and 2 minutes 30 seconds. Repeat this test for left bank. At no time during this test shall the pitch axis deviate from null by more than 1°.

4.7.9 Manual Fast Erection.- Drive the spin axis to 11° in right bank as described above. When the indicator reads 11° remove pins H, K, and V from S. Connect pin H to S. The time to erect from 10° to 5° shall be not more than 30 seconds. Remove pin H from S. Repeat for left bank. Repeat the above test for climb and dive, connecting pins H, K, and M to S to drive to 11°. Connect pin H to S to fast erect the spin axis.

4.7.10 Drift.- The Vertical Gyro shall be mounted on a roll, pitch, and yaw mechanism, with "Fore" pointed to the geographic north. With the Vertical Gyro in its normal mounting position and erected to vertical, connect pins V and M to S. The roll, pitch, and yaw mechanism shall be operated at a speed of 5 to 7 oscillations per minute with a total movement of 2-1/2°. With proper correction for the earth's rotation, the drift rate in either axis shall not exceed 0.4° per minute. This test shall be conducted for a five minute period and at the end of that time, the spin axis shall not have drifted more than 2° in a northerly or southerly direction, 1° in an easterly direction, or 3° in a westerly direction. Repeat the test with "Fore" pointed to the geographic south, east, and west.

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4.7.11 Vertical Erection Error (Pitch and Roll).- The gyro shall be operating at rated power and with normal voltage applied to the torque motors. The Vertical Gyro shall be mounted on a pitch, roll, and yaw test stand adjusted for 5 to 7 cycles per minute at a displacement of ±1-1/2°. Erect the gyro to established vertical, clockwise and counter-clockwise, about each axis from a displacement of approximately 3° to 5°. When erecting the gyro in pitch, both pitch and roll shall erect to established vertical to within a cone of half angle equal to 15 minutes of arc. When erecting the gyro in roll, both roll and pitch shall erect to established vertical within a cone of half angle equal to 15 minutes of arc.

4.7.12 Loop Test.- The Vertical Gyro shall be mounted on a fixture permitting rotation about the pitch axis. The unit shall be rotated one complete revolution (loop) at a rate of 10° ±1° per second, and returned to the normal position. The error in either pitch or bank shall not exceed 3°.

4.7.13 Power Interruption.- After a minimum of five minutes of operation, the power shall be removed (the circuit between T and U shall immediately open). The Vertical Gyro shall be moved through angles of ±60° in pitch and ±75° in roll and returned to level. Power shall be applied 40 seconds after removal. The pitch and bank axes shall indicate level within 1° and there shall be no recycling of the initial erection. The circuit between pins T and U shall immediately close.

4.7.14 System Vertical Gyro Tests.- The Vertical Gyro shall be subjected to system individual tests in accordance with Specification MIL-C-21013.

4.7.15 Voltage and Frequency Variation.- The varying voltage and frequencies shall be alternating current voltage at 105 to 125 volts and 380 to 420 cps. At any combination of the aforementioned voltage and frequency, the Vertical Gyro shall meet the requirements of the individual tests of this specification.

4.7.16 High Temperature Operation.- (This test may be combined with Section 4.7.19 High Temperature Exposure Test at the discretion of the testing agency.) The Vertical Gyro shall be tested in accordance with Procedure II of Specification MIL-E-5272 except that the exposure period shall be 4 hours. During the last 15 minutes of the 4 hour period, power shall be applied to the Vertical Gyro. At the end of the 4 hour period and with the temperature maintained at 71°C (160°F), the Vertical Gyro shall be subjected to and meet the requirements of Sections 4.7.4 Initial Erection, 4.7.5 Final Erection, 4.7.8 Erection Rate, and 4.7.10 Drift. The Vertical Gyro shall then be returned to standard conditions and allowed to stabilize for a period of 4 hours after which it shall be operated and meet the requirements of Sections 4.7.4 Initial Erection, 4.7.5 Final Erection, 4.7.8 Erection Rate, and 4.7.10 Drift. There shall be no damage which would adversely affect subsequent operation.

4.7.17 Low Temperature Operation.- (This test may be combined with the Section 4.7.20 Low Temperature Exposure Test at the discretion of the testing agency.) The Vertical Gyro shall be tested in accordance with Procedure I of Specification MIL-E-5272. The exposure period shall be 4 hours. During the last 15 minutes of the 4 hour period, power shall be applied to the Vertical Gyro. At the end of the 4 hour period and with the temperature maintained at -54°C (-65°F), the Vertical Gyro shall be subjected to and meet the requirements of the following Sections 4.7.4 Initial Erection, 4.7.5 Final Erection, 4.7.8 Erection Rate, and 4.7.10 Drift. The Vertical Gyro shall then be returned to standard conditions and allowed to stabilize for a period of 4 hours after which it shall be operated and shall meet the requirements of Sections 4.7.4 Initial Erection, 4.7.5 Final Erection, 4.7.8 Erection Rate, and 4.7.10 Drift. There shall be no damage which would adversely affect subsequent operation.

4.7.18 Immersion (Leakage) Test.- The Vertical Gyro shall be subjected to an Immersion (leakage) test in accordance with Procedure I of Specification MIL-E-5272.

4.7.19 High Temperature Exposure.- The Vertical Gyro shall be placed within the test chamber. The internal temperature of the chamber shall be raised to 93° ±2°C (200° ±3.6°F). The Vertical Gyro shall be exposed to this temperature for a period of 24 hours. The temperature of the chamber shall then be lowered to 71°C (160°F) and the Vertical Gyro shall be tested in accordance with Procedure II of Specification MIL-E-5272, except that the exposure period shall be 4 hours. During the last 15 minutes of the 4 hour period, power shall be applied to the Vertical Gyro. At the end of the 4 hour period and with the temperature maintained at 71°C (160°F), the Vertical Gyro shall be subjected to and meet the requirements of Section 4.7.4 Initial Erection,

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4.7.5 Final Erection, 4.7.8 Erection Rate, and 4.7.10 Drift. The Vertical Gyro shall then be returned to standard conditions and allowed to stabilize for a period of 4 hours after which it shall be operated and meet the requirements of Sections 4.7.4 Initial Erection, 4.7.5 Final Erection, 4.7.8 Erection Rate,and 4.7.10 Drift. There shall be no damage which would adversely affect subsequent performance.

4.7.20 Low Temperature Exposure.- The Vertical Gyro, shall be tested in accordance with Procedure II of Specification MIL-E-5272, except that the exposure periods shall be 24 hours at -62°C (-80°F) and 4 hours at -54°C (-65°F). During the last 15 minutes of the 4 hour period, power shall be applied to the Vertical Gyro. At the end of the 4 hour period and with the temperature maintained at -54°C (-65°F), the Vertical Gyro shall be subjected to and meet the requirements of the following sections 4.7.4 Initial Erection, 4.7.5 Final Erection, 4.7.8 Erection Rate, and 4.7.10 Drift. The Vertical Gyro shall then be returned to standard conditions and allowed to stabilize for a period of 4 hours after which it shall be operated and meet the requirements of Sections 4.7.4 Initial Erection, 4.7.5 Final Erection, 4.7.8 Erection Rate, and 4.7.10 Drift. There shall be no damage which would adversely affect subsequent performance.

4.7.21 Radio Noise Interference.- The electrical system of the Vertical Gyro shall be tested for conducted and radiated radio noise interference in accordance with Specification MIL-I-6181.

4.7.22 Vibration.- The Vertical Gyro shall be tested in accordance with Paragraphs 4.7.12, 4.7.12.1, and 4.7.12.2 of Procedure XII of Specification MIL-E-5272. The high temperature portion of the vibration test shall be conducted at 71°C (160°F). The Vertical Gyro shall be functioning in accordance with the requirements of this detail specification during the entire test period. Upon completion of the vibration period the Vertical Gyro shall be examined for failures and shall meet the requirements of Sections 4.7.4 Initial Erection, 4.7.5 Final Erection, 4.7.8 Erection Rate, and 4.7.10 Drift.

4.7.23 Impact Shock.- The Vertical Gyro installed in its shockmount and running shall not suffer damage or subsequently fail to meet the operational performance requirements when subjected to peak impact shocks of 10 ±1.0g with each shock impulse having a maximume time duration of 12 milliseconds. Six shock impulses shall be applied in a vertical direction relative to the normal upright position of the unit. The test equipment shall be a sand box and drop rig, or equivalent, in accordance with Specification MIL-S-4456. Upon completion of the test, the Vertical Gyro shall operate in accordance with the requirements of Sections 4.7.4 Initial Erection, 4.7.5 Final Erection, 4.7.8 Erection Rate, and 4.7.10 Drift. Inspection of the Vertical Gyro shall reveal no mechanical damage or functional weakness.

4.7.24 Life.- The Vertical Gyro shall be operated at random headings for a period of 1000 hours under the test conditions specified in TABLE I.

TABLE I

Conditions	Test Condition	Time (Approx. Hours)
1	3° (±1-1/2°) Total Angle Pitch, Roll, and Yaw Motion	300
2	15° (±7-1/2°) Total Angle, Pitch, Roll, and Yaw Motion	300
3	Vibration - 0.002 to 0.005 inch amplitude and a frequency of 1500 to 2000 vibrations per minute	300
4	Tests conducted in accordance with this specification as specified by the Inspector	100

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Nothing aforementioned shall be construed to indicate that Conditions 1, 2, 3, and 4 may not be distributed in any manner through the life test and that the total time under each condition has to be performed consecutively or without interruption. During this testing period, the Vertical Gyro shall comply with the requirements of Section 4.7.11 Vertical Erection Error (Pitch and Roll) at intervals of not more than 50 hours with the Vertical Gyro operating under Condition 1 prescribed in TABLE I. Upon completion of 1000 hours of testing operation, the Vertical Gyro, without adjustments during the aforegoing period, shall conform to the requirements of Sections 4.7.4 Initial Erection, 4.7.5 Final Erection, 4.7.8 Erection Rate, and 4.7.10 Drift.

4.7.25 Humidity.- The Vertical Gyro shall be tested in accordance with Procedure I of Specification MIL-E-5272, except that the total time of testing shall be 120 hours (5 cycles). At the conclusion of the 120 hour period, the equipment shall be returned to standard conditions. The Vertical Gyro shall then be operated within one hour and shall meet the requirements of Sections 4.7.4 Initial Erection, 4.7.5 Final Erection, 4.7.8 Erection Rate, and 4.7.10 Drift. Examination for deterioration shall be made between 24 and 48 hours after completion of the humidity exposure. There shall be no evidence of corrosion or other deterioration which will affect the subsequent operation of the Vertical Gyro.

4.7.26 Salt Spray.- The Vertical Gyro, with external connections made to simulate installed conditions, shall be tested in accordance with Procedure I of Specification MIL-E-5272, except that the test period shall be 50 hours. At the end of the 50 hour period, the Vertical Gyro shall be subjected to and meet the requirements specified in Sections 4.7.4 Initial Erection, 4.7.5 Final Erection, 4.7.8 Erection Rate, and 4.7.10 Drift. There shall be no evidence of corrosion or other deterioration which will affect subsequent operation of the Vertical Gyro.

5. PREPARATION FOR DELIVERY

5.1 Packaging.-

5.1.1 Level A.- The Vertical Gyro shall be packaged in accordance with Specification MIL-P-7936. The method of preservation shall be in accordance with Specification MIL-P-116, Method IId, using metal reusable containers.

5.1.2 Level C.- When this level is required, packaging shall be in accordance with standard commercial practice.

5.2 Packing.-

5.2.1 Levels A and B.- The Vertical Gyro shall be packed in accordance with Specification MIL-P-7936. The level or levels required shall be as specified in the invitation for bid or contract.

5.2.2 Level C.- The Vertical Gyro shall be packed to insure that the shipment arrives in a satisfactory condition at destination. The shipment shall conform to the applicable carriers rules and regulations in effect at the time of shipment.

5.3 Marking.- The interior and exterior containers shall be marked as specified in Specification MIL-P-7936.

5.3.1 Precautionary Marking.- The following precautionary marking shall appear on two opposite sides of each interior package and shipping container whenever practicable depending on the size of the carton:

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Intended Use.- The Vertical Gyro covered by this specification is a component of the MP-1 Compass, Directional Reference System which is intended for use in fighter, attack, patrol, transport, and cargo type aircraft. The range of operation will be world-wide, including polar regions, as well as all aircraft requiring medium precision heading reference stabilization. This system will provide directional information to remote indicators, electronic equipment, radar equipment, navigational

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equipment, and furnish directional control for autopilots, bomb directors, automatic approach equipment, and other auxiliary systems requiring directional reference intelligence. The main features of this new system are its precise directional accuracy, compactness, lightweight, reliability, flexibility of application, ease of installation and maintenance, compensation for magnetic deviation, correction for apparent gyro drift, and compass indication stability independent of latitude, as well as permitting the aircraft swing procedure to be accomplished rapidly and accurately.

6.2 Ordering Data.- Requisitions, contracts, and orders should state the MS part number of the Vertical Gyro desired, the quantity, and whether overseas packing is to be furnished.

6.3 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the bid opening date, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date.

6.3.1 The attention of suppliers is called to this requirement and manufacturers are urged to arrange to have the products, that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Bureau of Aeronautics, Department of the Navy, Washington 25, D.C., the qualifying agency.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEM PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM 1426, APR 63

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOL[ETE]

PLATE NO. 19419

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